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碩士論文

運用資料探勘技術協助專利維護決策

Supporting Patent Maintenance Decision: A Data Mining Approach

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摘要

專利的功能除了保護發明者的智慧心血不被侵害之外，對於企業在日漸激烈的商業環境上也愈顯重要。然而許多國家的專利制度，在專利申請獲准之後，必須要定期繳交維護費用，才能確保此專利在期限內仍然有作用。然而，並非所有的專利都需要進行維護。對於企業而言，找出「重要專利」進行維護，並將「不重要專利」進行放棄，這同時包含許多不同的決策思維。本研究預計透過德菲法(Delphi method)獲得實務上認為會影響專利是否維護的因素，且運用資料探勘的方式建立系統支援專利維護決策。這種以資料探勘為基礎的專利維護決策系統，可以協助組織在進行維護決策時，增進其效率且降低維護決策的成本。實證結果顯示，我們所提出的維護模型表現優於基準模型(所使用的變數僅由文獻中所建議組成）。

關鍵字：專利維護、資料探勘、專利分析

Abstract

Protecting the rights of inventor is not the only purpose of patents, and it is becoming more and more important under the keen competitive business environment nowadays. In the patent systems of many countries, patent holders are required to pay a maintenance fee after the initial application to retain patent protection on any invention until the expiration of the protection period. However, not all patents are worth maintaining by patent holders. Thus, organizations need to identify "important patents" for maintenance and abandon "unimportant patents." In this study, we attempt to examine factors influencing patent maintenance via a Delphi study and propose a data mining approach to support patent maintenance decision. Such a data-mining-based patent maintenance decision support system can help organizations improve the effectiveness of patent maintenance decisions and, at the same time, decrease the cost of maintenance decision. Our empirical evaluation results suggest that our proposed model outperforms the benchmark model (i.e., involving the variables suggested by the literature only).

Keywords: Patent maintenance, Patent renewal, Data mining, Patent analysis

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2010 夏 清大台積館

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Chapter 1

INTRODUCTION

1.1 Background

The role of intellectual assets (IA) is becoming more and more significant in the era of knowledge economy we are in now. IA includes traditional assets such as patents, copyrights, trademarks, trade secrets, and domain names, as well as certain tacit knowledge like know-how, business practices and customer preferences (Tao et al., 2005). Compare to other IAs, patent is more tangible to measure, and the importance of patent is much more significant in terms of commercial pursuits and technology.

The basic function of patents is to provide the granted right to an inventor for a certain period of time (USPTO). Protecting the rights of inventor is not the only purpose of patents. Its significance in technology management is also vital, for instance, knowledge management, R&D portfolio management, strategic planning, human resource management, and merger and acquisition decision support (Ernst, 2003).

In many countries, patent holders are required to pay a maintenance fee after the initial application to retain patent protection on any invention until the expiration of the protection period. In the United States, the term of protection is 20 years from application, and it's necessary for the patent holder to pay a maintenance fee every four years. Furthermore, the maintenance fee increases over the years. The charging criterion for a large entity is US$980 in between the 3.5th and the 4th year, $2,480 in between the 7.5th and the 8th year, and $4,110 in the final 11.5th and the 12th year. **Table 1.1** shows the details of maintenance fees in U.S. – a small entity in U.S. needs

to pay $3,785 to ensure a full-term legal right and a large entity has to pay $7,570 to secure the same result.

Table 1.1 Maintenance Fees in U.S

	Due at 3.5 years	Due at 7.5 years	Due at 11.5 years
Small Entity[1]	$490	$1240	$2055
Large Entity	$980	$2480	$4110

Source: USPTO

The patent renewal rate from the USPTO 2009 Performance and Accountability Report is shown in



[1] Small Entity: Person, Small business concern, Nonprofit Organization, License to a Federal agency, Security Interest

Table 1.2. It shows the percentage of patent renewal rate in the most recent five years. For example, in 2009, 80.3% of the patents issued three years earlier were renewed, 63.5% of the patents issued seven years earlier were renewed, and 45.4% of the patents issued 11 years earlier were renewed.

On the other hand, the maintenance rate could differ from industry to industry. In countries like Finland and Norway, prior studies show that the maintenance rate is higher in chemical industry than in heavy industry and low-tech industry (Pakes et al., 1989). In a 1991 U.S. study, patents in the category of “Other Industries” had the lowest renewal rate of 16.34% while patents in the industry of Computer and Communications had the highest renewal rate of 53.24% (Bessen, 2008).

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Table 1.2 Patent Renewal Rates

Patent Renewal Rates[2]	2005	2006	2007	2008	2009
First Stage	83.1%	93.1%	90.1%	83.1%	80.3%
Second Stage	65.4%	69.2%	71.4%	73.7%	65.3%
Third Stage	45.0%	44.4%	48.5%	49.2%	45.4%

Source: 2009 Performance and Accountability Report

1.2 Research Motivation and Objective

There are several reasons that would influence the patent maintenance decision. The context of patent maintenance provides an interesting window through which we could explore how the patent holder weighs the "important" or "unimportant" factors in determining whether to maintain a patent or not.

In general, the decision of patent renewal might be influenced by the high cost of the maintenance fee (Griliches, 1990) or the limited resources of the firms. Patent holder thinks some patents might not bring revenue in the future, or the expected expenditure might exceed the expected income, so eliminating these patents would reduce unnecessary costs of maintenance fee. Some companies take patent maintenance as knowledge assets, and it would effectively improve their return. Dow Chemical saved $4 million during the first year of its knowledge management initiative and it is expecting to generate more than $100 million in licensing revenues in the future that

[2] the First Stage refers to the end of the 3rd year after the initial patent is issued; the Second Stage refers to the end of the 7th year after the initial patent is issued; and the Third Stage refers to the end of the 11th year after the initial patent is issued.

might otherwise have forgone. Companies such as IBM (with 3,000% increase in revenues from licensing its intellectual capital), Microsoft, and Texas Instruments demonstrated the power of licensing to improve the financial performance by managing and carefully reviewing the patents (Jones et al., 2002).

In addition, another reason that companies make the maintenance decision is to gain protections from patent right. Patent become a competitive tool with strategic function, litigations are more common nowadays. Some patents are maintained in order to avoid the company from losing money from litigations and gain advantage in negotiations (Crampes & Langinier, 1998; Pakes, 1986; Schankerman & Pakes, 1986). On the other hand, consider the strategy in combination with R&D and knowledge management and so on. To determine whether the patent is worth maintenance, the company has to look beyond one single patent but to take the entire relations of all patents into consideration. The renewal probability is higher for patents that belong to internal sequential innovation than patents of stand-alone innovation (Liu et al., 2008).

The patent law offices take the responsibility to determine whether a patent should be maintained in businesses, and there are no universal guidelines yet. This task consumes a large amount of time and money mainly due to the reason that it requires the work of a lot of people.

The maintenance factors mentioned above are only part of the overall consideration. In this study, we attempt to examine factors influencing patent maintenance and propose a data mining method to support patent maintenance decision. Such a data-mining-based patent maintenance decision support system can help organizations improve the effectiveness of patent maintenance decisions and, at the same time,

decrease the cost of maintenance decision.

This chapter discusses the research background, motivation and objective of this study. The remainder of this thesis is organized as follows. Chapter 2 reviews literature related to this study. In Chapter 3, we depict our Delphi study to find the dimensions that influence the decision of patent maintenance. In Chapter 4, we describe our design of the data-mining-based patent maintenance decision support system. Subsequently, Chapter 5 reports our empirical evaluations, including the data collection, the experiment design and important evaluation results. Finally, we conclude in Chapter 6 with a summary as well as some future research directions.



Chapter 2

LITERATURE REVIEW

In this chapter, we review the existing studies related to patent maintenance. The characteristics and variables of full-term patents compared with non-maintained patents are discussed below. These previous studies have demonstrated some criteria that help determine whether a patent should be maintained.

2.1 Factors Influence Patent Maintenance Decision

- **Economy Aspect**

Prior studies show that economical factors are essential to patent maintenance decision. These studies assume that patent holder makes maintenance decisions based on the revenue from the patent and the maintenance fee. In other words, the patent holder decides not to pay the maintenance fee when the expected benefits from the patent are less than its costs (Crampes & Langinier, 1998; Pakes, 1986; Schankerman & Pakes, 1986).

- **Technology Aspect**

Patent maintenance decision should support the technology portfolio of an organization. A patent may be very important in subsequent technological innovations, but this not necessarily means that it will bring profits for the assignee. Biases might appear if only economic factor is considered; technology factor along with the firm's strategies are also crucial in terms of making patent maintenance decisions. In other analyses of patent renewal data, the technological aspect on patents may have a

greater influence upon renewal decisions than economic concerns regarding the costs of their renewal (Thomas, 1999).

2.2 Characteristics of Patent Renewals

Prior studies show that full-term renewed patents as compared to those abandoned patents have certain characteristics:

- ***Backward citation***: more (Harhoff et al., 2003; Moore, 2005)
- ***Claim***: more (Barney, 2003; Moore, 2005)
- ***Claim length***: shorter (Barney, 2003)
- ***Family size***: larger (Harhoff et al., 2003; Lanjouw et al., 1998)
- ***Forward citation***: more (Barney, 2003; Harhoff et al., 1999; Harhoff et al., 2003; Thomas, 1999)
- ***Length of written specification***: longer (Barney, 2003)
- ***Patent scope***: higher (Harhoff et al., 2003)
- ***Science linkage***: more (Harhoff et al., 2003)
- ***Self citation***: more (Thomas, 1999)
- ***Patent sequence***: yes (Liu et al., 2008)

Previous studies on full-term patents and variables are not enough, because they examine the characteristics of full-term patents with only one or few variables.

Chapter 3

DELPHI STUDY

In this research, we try to find the dimensions that affect patent maintenance by the Delphi method. The Delphi method is to invite a group of experts to solve complex problems, and some characteristics of the method include structuring of information flow, regular feedback and ensured anonymity of the participants (Dueker, 1996). We use this method to get a wider and deeper survey about the factors affecting patent maintenance.

We take three rounds in Delphi method, because the review suggests that too many rounds may decrease the effect (Dueker, 1996). The expert group is formed by five directors who have worked in patent & trademark law offices for at least 7 years (average 16.2 years) with previous experience in patent related fields (**Table 3.1**).

Table 3.1 Summary of Experts Participating in Our Delphi Study

Number	Current Occupation	Years in patent related fields
1	Patent Attorney	30
2	President / Representative Attorney	14
3	Patent Manager	7
4	Patent Manager	10
5	Patent Attorney	20

3.1 First Round

We conduct a pilot test and adjust the phrasing before the first round. The questionnaire sent out in the first round to the participating experts is as follows: "*After the patent is granted, and patent holder needs to pay maintenance fee to retain patent protection. But it is not necessary for firms to maintain all patents. You need to give up some patents due to certain reasons, what is your criterion to determine whether to maintain the patent or not?" Please write down the factors that you think might be considered and please describe each factor*."

After the first round, we collect all factors identified by all participating experts. The average number of factors from each expert is 8. Because these experts did not discuss with each other prior to the survey and it is possible that some factors from different experts might duplicate, we deem necessary to establish a mechanism to deal with such duplications. One coder examines all factors first and provides a consolidated list of factors from all experts. Specifically, when factors with the same or semantically similar description appear in different names from different experts, they will be combined into one factor. Subsequently, the consolidated list of factors is reviewed by two reviewers, one is a senior patent manager in a high-tech company who has been practicing patent related fields for 10 years and another is an academic research with at least 4 years of experience in patent analysis and patent mining. They review the consolidated list individually and meet together to resolve any concern identified by either reviewer, and then determine the final consolidated list of factors. As a result, we combine 10 factors with identical or semantically similar descriptions and the consolidated result contains 30 factors. In addition, according to the meanings of the factors, we construct 11 dimensions that cover these factors.

3.2 Second Round

In the second round, we list the original question along with the consolidated factors but do not reveal the names of the experts. Next, the experts are asked to mark the factors they think is relevant and delete those factors they consider irrelevant. At the same time, the experts could still add new factors while they see the factors proposed by all experts.

In this round, the average number of factors an expert deem relevant is 27.6. In other words, the experts think the number of factors irrelevant to patent maintenance is 2.4. We removed the factors that over half of the experts consider irrelevant to patent maintenance, and we also found that 24 of all the factors are deemed relevant by all the experts, and it shows that the opinions of these experts in patent maintenance share certain similarities

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3.3 Third Round

The process in the third round is the same as that in the second round. We mark the percentage each factor has been brought up along with the number of times the factor's ability to affect maintenance decisions has been agreed and disagreed on. According to these information, experts are asked to mark the factors they think is relevant or not in patent maintenance decision. As in the second round, the experts could still add new factors while they see factors proposed by all experts.

In the third round, we found that the opinions from the experts have not changed, and it suggests that the result has already reached a high level of convergence. After three rounds, we obtain the dimensions that influence patent maintenance decision (see

Table 3.2).

Table 3.2 Dimensions and Factors Affecting Patent Maintenance Decision

Dimension	Brief Description
Business Strategy	Direction of the company's business, marketing strategies, and future competitiveness
Patent Use	Use for production, and probability of future use
Commercialization	Whether it is utilized or embedded in a final product, and the possibility of the patented technology being commercialized
Market Scale	Maturity of market
Technology Transfer & Licensing	Possibility of patent licensing, and probability of future transfer
Patent Family	Whether the patent belongs to core patent family, whether some other patents are extended from the patent, and whether the patent is applied in multiple countries
Patent Quality	Whether the patent is influenced by prior arts
Patent Defensiveness	Scope of the claims, difficulty of design around, possibility of preventing litigations from patent troll.
Technology Type	Technology lifecycle of patent, whether the patent belongs to part of the technology standard, how advance the technology protected by the patent is, whether the technology protected by the patent is a core technology
Future R&D	Possibility of future innovation extended from the patent
Patent Age	Remaining life of the patent

Chapter 4

DESIGN OF PATENT MAINTENANCE DECISION SUPPORT SYSTEM

As mentioned, we attempt to use data mining techniques to develop a patent maintenance decision support system. In this study, we formulate patent maintenance decision as a classification problem; i.e., classifying whether a patent should be "maintained" or "abandoned". To construct this model, we first have to identify variables reflecting these dimensions that influence patent maintenance decisions, and then to construct a classification model on the basis of a set of training instances.

4.1 Variables Representing Dimensions

The dimensions are the main concept of patent maintenance; however our model needs the numeric or nominal data, so these dimensions cannot be used in the data mining system directly. In this step we identify variables that reflect the specific dimensions identified in our Delphi study.

There are two main ways to define the variables: one is from literature review and the other is to design by ourselves. Through the description from the Delphi study, we match the dimensions with related variables such as citation, claims, family size, and so on. Prior studies show that these variables are related to patent maintenance. For those factors that do not match the literature review, we will design variables according to the existing data sources (e.g., USPTO, EPO) to represent each of these dimensions.

4.1.1 Business Strategy

The business strategy includes financial factor, market strategy, etc, and it is hard to be accessed directly. The classification codes (IPC, USPC) can represent different technologies, and it is also the factor influencing the decision of patent maintenance (Thomas, 1999).

Compatibility of Technology Scope

Hence, through the classification codes, we can understand the research direction of a company. For the maintenance decision by the same assignee last year, we can tell which classification codes were maintained and which were not maintained.

Maintenance compatibility of technology scope (by IPC)

$$= \frac{\sum_{i=1}^{m} \frac{|X \cap M_i|}{\sqrt{|X|}\sqrt{|M_i|}}}{m}$$

$X =$ the IPC code of current patent

$M_i =$ the IPC code of maintained patent i with same assignee last year

$m =$ the number of maintained patent with same assignee last year

Abandonment compatibility of technology scope (by IPC)

$$= \frac{\sum_{i=1}^{a} \frac{|X \cap A_i|}{\sqrt{|X|}\sqrt{|A_i|}}}{a}$$

$X =$ the IPC code of current patent

$A_i =$ the IPC code of abandoned patent i with same assignee last year

$a =$ the number of maintained patent with same assignee last year

Compatibility of technology scope (by IPC)

$=$ Maintenance compatibility of technology scope

$-$ Abandonment compatibility of technology scope (by IPC)

Maintenance compatibility of technology scope (by USPC)

$$= \frac{\sum_{i=1}^{m} \frac{|X \cap M_i|}{\sqrt{|X|}\sqrt{|M_i|}}}{m}$$

$X =$ the USPC code of current patent

$M_i =$ the USPC code of maintained patent i with same assignee last year

$m =$ the number of maintained patent with same assignee last year

Abandonment compatibility of technology scope (by USPC)

$$= \frac{\sum_{i=1}^{a} \frac{|X \cap A_i|}{\sqrt{|X|}\sqrt{|A_i|}}}{a}$$

$X =$ the USPC code of current patent

$A_i =$ the USPC code of abandoned patent i with same assignee last year

$a =$ the number of maintained patent with same assignee last year

Compatibility of technology scope (by USPC)

$=$ Maintenance compatibility of technology scope

$-$ Abandonment compatibility of technology scope (by USPC)

4.1.2 Commercialization

Commercialization is where a patent is utilized or embedded in a final product and the possibility of the patented technology being commercialized, the wider the patent

scope is, the higher the commercialization will get.

Patent Scope

Measuring the patent scope is difficult, so we follow prior studies by using the technology classification codes to compute (Harhoff et al., 2003). If an invention is wider and more original, the corresponding patent will be associated with more classification codes (Lanjouw et al., 1998).

Patent scope (by IPC)

= the number of different four-digit IPC classification codes of patent

Patent scope (by USPC)

= the number of different three-digit USPC classification codes of patent

4.1.3 Technology Transfer & Licensing

Transfer of Patent Assignee

In the real world, it is hard to know the possibility of technology transfer and licensing. However, we know the prior history of transfers of a patent and assume that if the focal patent has been transferred from another assignee to the target company, the company has a stronger motivation to maintain this patent.

Number of patent assignee transfer

= the number of patent assignee transfer

4.1.4 Patent Family

This dimension consider whether the focal patent belongs to core patent family of the target company, whether some other patents are extended from the focal patent, and

whether the patent is applied in multiple countries. We thus measure this dimension using two patent indicators: patent family and citations.

Patent Family

The definition of patent family is a group of patents protecting a particular invention in multiple countries. The number of years a patent is maintained and the patent family size are highly correlated (Harhoff et al., 2003; Lanjouw et al., 1998).

Patent family size (applications)

= the number of applications the patent protecting a particular invention

Patent family size (countries)

= the number of unique countries that are in the family

Major market

= the number of applications in three major patent offices (USPTO, EPO, and JPO)

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Citations

Backward citation is the reference to prior technology, either patents or other scientific literature on which the current patent builds on. Compared to the expired patent, full-term patents are considered more frequently cited, and it would generate higher economic value (Harhoff et al., 1999; Harhoff et al., 2003).

Number of backward citations

= the number of patents cited the particular patent

Science linkage index

= the number of other reference

Forward citation could measure the technology impact of a patent. More forward citations to the focal patent show how wide the relative technology is, and is also positively correlated with economic value (Barney, 2003; Maurseth, 2002; Moore, 2005; Thomas, 1999). However, forward citation would be affected by the age of the patent; it means that the older patent may receive more citations than younger patent just because of time. Therefore, we use the **Number of forward citations within 3 years**, **Number of forward citations within 5 years**, and **Average number of forward citations** to reduce the impact of time.

Number of forward citations

= the number of patents citing the particular patent

Number of forward citations within 3 years

= the number of patents citing the particular patent within 3 years after granted

Number of forward citations within 5 years

= the number of patents citing the particular patent within 5 years after granted

Average number of forward citations

$$= \frac{\text{the number of patents citing the particular patent}}{\text{the patent age(maintain event date} - \text{granted date)}}$$

4.1.5 Patent Defensiveness

The dimension of patent defensiveness includes the scope of the claims of the focal patent, the difficulty of design around, and the possibility of preventing litigations from patent trolls.

Claims

Claims in a patent document defined the scope of protection for the patented invention. Prior studies indicate that more claims broaden the scope of protection, and with the increase of independent claims maintenance rates also increase. The length of patents negatively influence the scope of technology (i.e., the more detail involved in describing the claims, the narrower the scope could get) (Barney, 2003; Moore, 2005).

Number of claims

= the number of claims in the patent

Number of independent claims

= the number of claims

− the number of dependent claims (claims with the term "claim")

Average length of independent claims

= the average length of independent claims

Length of description

= the word length of description in the patent

Approval Time

The approval time would be influenced by application history; prior study shows that the valid patent had significantly higher approval time than invalid patent (Cockburn et al., 2002). Approvals take longer times for more complicated patents.

Approval time

= granted date − applied date

4.1.6 Technology Type

This dimension intends to reflect the technology lifecycle of the focal patent, whether

the patent belongs to part of the technology standard, how advance the patented technology is, and whether the patented technology is a core technology.

Originality

The concept of the originality index is the how diverse a patent has been cited in the technology scope. When the focal patent made backward citation in different technology area, the originality is higher (Hall et al., 2001).

Originality index (by IPC)

$$= 1 - \sum_{c} \left(\frac{\text{the number of backward citations with current IPC c}}{\text{the number of backward citations}} \right)^2$$

Originality index (by USPC)

$$= 1 - \sum_{c} \left(\frac{\text{the number of backward citations with current USPC c}}{\text{the number of backward citations}} \right)^2$$

Generality

Similarly, the definition of generality index is the how diversely a patent has been received in technology scope. The power of influence is higher when the focal patent is cited by different technology areas, and the higher generality is (Hall et al., 2001).

Generality index (by IPC)

$$= 1 - \sum_{c} \left(\frac{\text{the number of forward citations with current IPC c}}{\text{the number of forward citations}} \right)^2$$

Generality index (by USPC)

$$= 1 - \sum_{c} \left(\frac{\text{the number of forward citations with current USPC c}}{\text{the number of forward citations}} \right)^2$$

Technology Life Cycle

Different technologies have different life cycles, and it influenced the patent maintenance decision. For patents with the same age, they might be abandoned in rapidly-changing industries such as 3C industry, but they may still be maintained in other industries. The concept of "cited half life" indicates the speed of technology advancement and the newness of patent, and we use the patent age of mean and median to compute. Besides, we design another variable to evaluate the life cycle of a patented technology. Specifically, it considers the maintenance status of the backward citations of the focal patent. When the value of this variable is higher, the possibility of maintaining the focal patent would be higher.

Cited half life (by mean)

= the mean patent age of backward citations

Cited half life (by median)

= the median patent age of backward citations

Maintenance of backward citations

$$= \frac{0 \times \sum E_1 + 1 \times \sum E_2 + 2 \times \sum E_3 + 3 \times \sum M}{\text{the number of backward citations}}$$

(Ei： the number of patent expired at i stage,

M： the number of patent maintained at all stage)

Cross State Cooperation

The patents made under a cross border cooperation structure have a higher average quality, and impacts of international cooperation are higher than domestic cooperation (Guellec & van Pottelsberghe de la Potterie, 2000).

Number of cross state cooperation

= the number of different states in which the inventors belongs

4.1.7 Future R&D

This dimension intends to reflect the possibility of future innovation extended from the focal patent.

Internal Sequential Innovation

Compared to stand-alone patents, a patent that belongs to a sequence of patented innovations is more likely to be maintained. The internal sequential innovations is defined as "internal sequence of innovations that build upon the same underlying technological trajectory and therefore share some genealogical connections"(Liu et al., 2008)

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Internal sequential innovation

= 1 if the patent belongs to sequence innovation (Division, Continuation, CIP) or 0 otherwise.

Self Citation

It means the focal patent's forward citations received or backward citations made from the same owner firms, and the higher self citations represent that the patent gets more attention by the firm. Self citation can be the proxy of subsequent invention in a company.

Forward self citation rate

$$= \frac{\text{the number of forward citations with same assignee}}{\text{the number of forward citations}}$$

Forward self citation rate within 3 years

$$= \frac{\text{the number of forward citations with same assignee within 3 years}}{\text{the number of forward citations within 3 years}}$$

Forward self citation rate within 5 years

$$= \frac{\text{the number of forward citations with same assignee within 5 years}}{\text{the number of forward citations within 5 years}}$$

Backward self citation rate

$$= \frac{\text{the number of backward citations with same assignee}}{\text{the number of backward citations}}$$

Table 4.1 lists all variables employed in our proposed patent maintenance decision support system.



Table 4.1 Summary of Variables involved in Our Study

Dimensions	Variables	Definition or Formula
Business Strategy	Maintenance compatibility of technology scope (by IPC)	$\frac{\sum_{i=1}^{m} \frac{\lvert X \cap M_i \rvert}{\sqrt{\lvert X \rvert}\sqrt{\lvert M_i \rvert}}}{m}$ $X =$ the IPC code of current patent $M_i =$ the IPC code of maintained patent i with same assignee last year $m =$ # of maintained patent with same assignee last year
	Abandonment compatibility of technology scope (by IPC)	$\frac{\sum_{i=1}^{a} \frac{\lvert X \cap A_i \rvert}{\sqrt{\lvert X \rvert}\sqrt{\lvert A_i \rvert}}}{a}$ $X =$ the IPC code of current patent $A_i =$ the IPC code of abandoned patent i with same assignee last year $a =$ # of abandoned patent with same assignee last year
	Compatibility of technology scope (by IPC)	= Maintenance compatibility of technology scope − Abandonment compatibility of technology scope (by IPC)
	Maintenance compatibility of technology scope (by USPC)	$\frac{\sum_{i=1}^{m} \frac{\lvert X \cap M_i \rvert}{\sqrt{\lvert X \rvert}\sqrt{\lvert M_i \rvert}}}{m}$ $X =$ the USPC code of current patent $M_i =$ the USPC code of maintained patent i with same assignee last year $m =$ # of maintained patent with same assignee last year
	Abandonment compatibility of technology scope (by USPC)	$\frac{\sum_{i=1}^{a} \frac{\lvert X \cap A_i \rvert}{\sqrt{\lvert X \rvert}\sqrt{\lvert A_i \rvert}}}{a}$ $X =$ the USPC code of current patent $A_i =$ the USPC code of abandoned patent i with same assignee last year $a =$ # of abandoned patent with same assignee last year

Dimensions	Variables	Definition or Formula
	Compatibility of technology scope (by USPC)	= Maintenance compatibility of technology scope – Abandonment compatibility of technology scope (by USPC)
Commercialization	Patent scope (by IPC)	# of different four-digit IPC classification codes of patent
	Patent scope (by USPC)	# of different three-digit USPC classification codes of patent
Technology Transfer & Licensing	Number of patents assignee transfer	# of patent assignee transfer
Patent Family	Patent family size (applications)	# of applications the patent protecting a particular invention
	Patent family size (countries)	# of unique countries that are in the family
	Major market	# of applications in three major patent offices (USPTO, EPO, and JPO)
	Number of backward citations	# of patents cited the particular patent
	Science linkage index	# of other reference
	Number of forward citations	# of patents citing the particular patent
	Number of forward citations within 3 years	# of patents citing the particular patent within 3 years after granted

Dimensions	Variables	Definition or Formula
	Number of forward citations within 5 years	# of patents citing the particular patent within 5 years after granted
	Average number of forward citations	$= \frac{\text{\# of patents citing the particular patent}}{\text{the patent age(maintain event date} - \text{granted date)}}$
Patent Defensive	Number of claims	# of claims in the patent
	Number of independent claims	= # of claims – # of dependent claims (claims with the term "claim" in their description)
	Average length of independent claims	= the average length of independent claims
	Approval time	= granted date – applied date
	Length of description	= the word length of description in the patent
Technology type	Originality index (by IPC)	$= 1 - \sum_{c} \left(\frac{\text{\# of backward citations with current IPC c}}{\text{\# of backward citations}} \right)^2$
	Originality index (by USPC)	$= 1 - \sum_{c} \left(\frac{\text{\# of backward citations with current USPC c}}{\text{\# of backward citations}} \right)^2$
	Generality index (by IPC)	$= 1 - \sum_{c} \left(\frac{\text{\# of forward citations with current IPC c}}{\text{\# of forward citations}} \right)^2$
	Generality index (by USPC)	$= 1 - \sum_{c} \left(\frac{\text{\# of forward citations with current USPC c}}{\text{\# of forward citations}} \right)^2$
	Cited half life (by mean)	= the mean patent age of backward citations

Dimensions	Variables	Definition or Formula
	Cited half life (by median)	= the median patent age of backward citations
	Maintenance of backward citations	$= \frac{0 \times \sum E_1 + 1 \times \sum E_2 + 2 \times \sum E_3 + 3 \times \sum M}{\#\text{ of backward citations}}$ (E_i : # of patent expired at i stage, M : # of patent maintained at all stage)
	Number of cross state cooperation	# of different states in which the inventors belongs
Future R&D	Internal sequential innovation	= 1 if the patent belongs to sequence innovation (Division, Continuation, CIP) or 0 otherwise.
	Forward self citation rate	$= \frac{\#\text{ of forward citations with same assignee}}{\#\text{ of forward citations}}$
	Forward self citation rate within 3 years	$= \frac{\#\text{ of forward citations with same assignee within 3 years}}{\#\text{ of forward citations within 3 years}}$
	Forward self citation rate within 5 years	$= \frac{\#\text{of forward citations with same assignee within 5 years}}{\#\text{ of forward citations within 5 years}}$
	Backward self citation rate	$= \frac{\#\text{ of backward citations with same assignee}}{\#\text{ of backward citations}}$

4.2 Process of Patent Maintenance Decision Support System

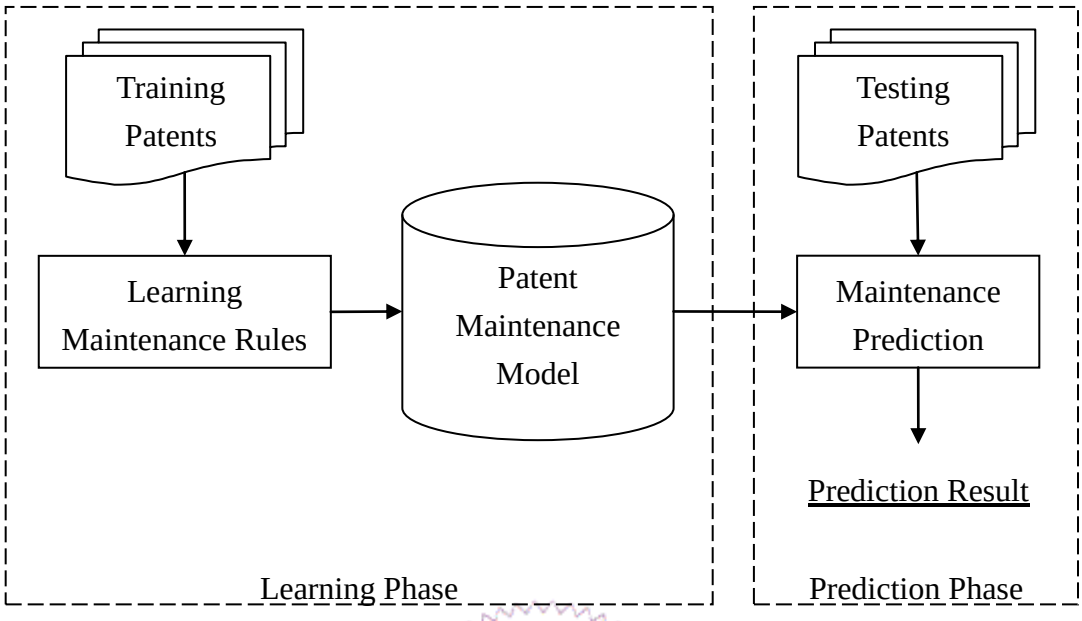


Figure 4.1 Overall Process of Patent Maintenace Decision Support System

Due to the fact that our proposed patent maintenance decision support system is constructed by the data mining approach, the system can be separated into two parts: leaning phase and prediction phase (see **Figure 4.1**). In the learning phase, we use training patents to learn a patent maintenance model with respect to the variables listed in Section 4.1. In the prediction phase, when the company faces the decision of patent maintenance and this model can provide the suggestion of the prediction result.

4.3 Learning Algorithms

In this study, we choose C4.5 decision tree and SVM (Support Vector Machines) as the underlying learning algorithms, and use the open source data mining software Weka to build a classifier, respectively. In order to achieve better performance, we use bootstrap aggregating (bagging) algorithm to improve the classification effectiveness.

Bagging algorithm is proposed by Breiman (1966), which is a method for generating multiple versions of a classifier and using these to get a combined prediction. At first, sampling *k* subsets from the original dataset with replacement, and building a classifier through the same underlying algorithm (e.g., C4.5 or SVM in our study) for each subset. Then, for a target prediction task, the predictions made by each base classifier is then combined by using the non-weighted voting method to arrive at an overall prediction (see **Figure 4.2**). In general, the selection of an odd number as *k* would avoid a tie, so we choose $k = 15$ in our study.

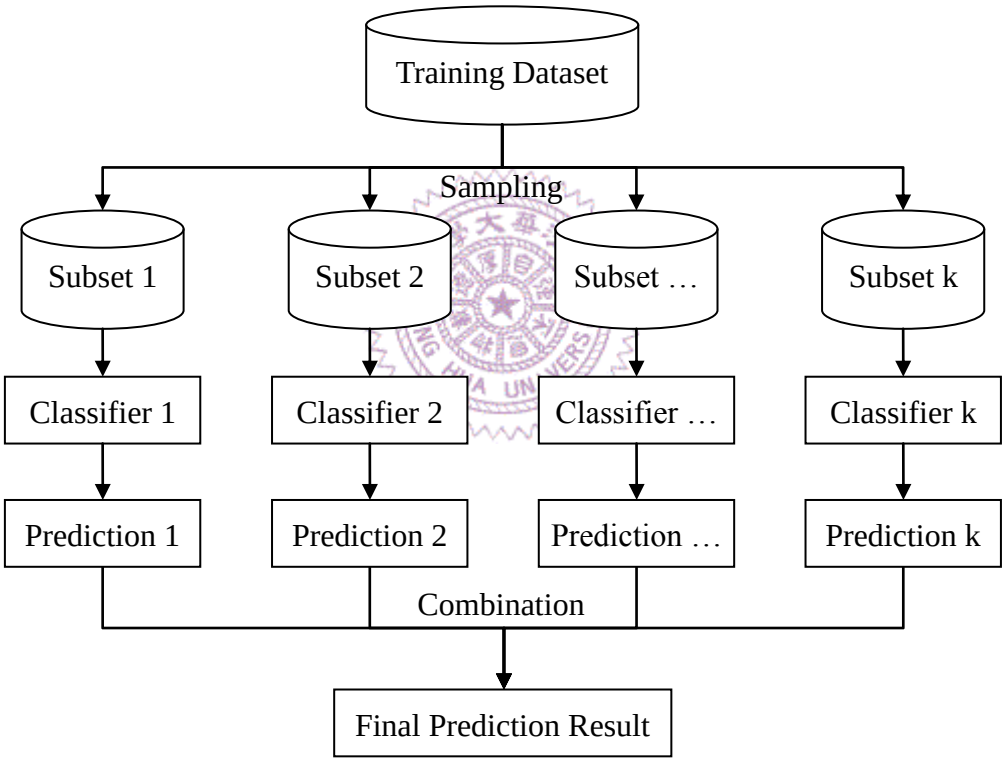


Figure 4.2 Overall Process of Bagging Algorithm

Chapter 5

EMPIRICAL EVALUATION

5.1 Data Collection

The first step of data collection is to choose industries and companies to be included in our evaluation. Specifically, we select industry sectors from Future Global 500[3] in 2005. There are 50 sectors in total; however, some industries have fewer patents such as Banks, Insurance, and Wholesalers. Therefore, we exclude these industries. Furthermore, for those industries with less than 3 companies listed, we consider them lack of representativeness and thus also exclude them from our consideration. As a result, we select 7 industries in our empirical evaluation. They include *Aerospace and Defense*, *Chemicals*, *Electronics*, *Electrical Equipment*, *Motor Vehicles & Parts*, *Network and Other Communications Equipment*, and *Pharmaceuticals*. In these industries, there are 34 companies and 20,813 patents in the third stage that have faced maintenance decision during 2003 to 2006 (see **Table 5.1**). Because some variables depend on the patent maintenance status of the respective company of the prior year, we cannot include patent maintenance instances in 2003 in our dataset. As a result, we only use the patent maintenance instances during 2004 and 2006 in our evaluation (i.e., a total of 16,205 patents are included).

[3] http://money.cnn.com/magazines/fortune/global500/2005/index.html

Table 5.1 Profiles of Industries and Companies Included in Our Evaluation

Industry	Number of Companies	Number of Patents
Aerospace and Defense	5	1775
Chemicals	6	2829
Electronics, Electrical Equipment	3	6720
Motor Vehicles & Parts	6	2864
Network and Other Communications Equipment	4	2885
Pharmaceuticals	6	1213
Telecommunications	4	2527
Sum	34	20813

Table 5.2 Maintenance Rates of Third Stage

Industry	2003	2004	2005	2006
Aerospace and Defense	84.63%	83.75%	83.98%	85.98%
Chemicals	62.04%	60.17%	58.02%	60.47%
Electronics, Electrical Equipment	69.57%	75.18%	68.57%	71.11%
Motor Vehicles & Parts	69.58%	68.31%	60.59%	71.33%
Network and Other Communications Equipment	80.93%	74.10%	70.33%	73.63%
Pharmaceuticals	81.38%	69.79%	63.58%	56.27%
Telecommunications	77.28%	80.67%	87.27%	90.21%

In the second step, we obtain the maintained record of each patent in our dataset from USPTO[4]. The data retrieved contain patent number, maintenance fee event date, and maintenance fee events. There are total 121 maintenance event codes, and we only focus on the maintenance decisions in the third stage. So the status of "payment of maintenance fee, 12th Year" means the patent is maintained and the "Patent Expired for Failure to Pay Maintenance Fees" means the patent was abandoned. The maintenance rates across different industries in our dataset are shown in **Table 5.2**.

The third step is to extract variables involved in our classification model from online sources for each patent in our dataset. We use special java machine to retrieve the USPTO patent full-text database, and compute values of these variables for each patent.

5.2 Cross Validation

To evaluate the effectiveness of our proposed patent maintenance decision support system, a ten-fold cross validation is employed. Thus, all the instances in our dataset (i.e., 16,205 patents) are randomly partitioned into ten mutually exclusive subsets of nearly equal size. A single subset is retained as the testing set for evaluating the system, and the other subsets are used as the training set. The process is then repeated 10 times, with each of the subsets used once as the testing set.

We examine and compare the effectiveness of our model in terms of accuracy, recall, precision rates and F1 Measure. **Table 5.3** shows the cross validation on our patent maintenance model, and we can find that for most performance indicators, C4.5 is

[4] https://eipweb.uspto.gov/MaintFeeEvents/

better than SVM. In addition, the use of bagging algorithm achieves a higher effectiveness than that without bagging.

The breakdown accuracy of each industry is shown in **Table 5.4**. The accuracy achieved by the four algorithms investigated (i.e., SVM, C4.5, Bagging-SVM, and Bagging-C4.5) is all higher than the actual maintenance rate in some industries such as *Motor Vehicles & Parts*, *Network and Other Communications Equipment*, and *Pharmaceuticals*. On the other hand, the accuracy of all the algorithms is lower than the actual maintenance rate in *Aerospace and Defense* and *Telecommunications*. A plausible reason is that the classification performance may be influenced by the imbalanced class distribution between the "maintain" (i.e., positive) and "abandon" (i.e., negative) classes. The original dataset is asymmetric (Y: N = 11642: 4563), and the class distribution in each industry is different.

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Table 5.3 Cross Validation Results (with the Original Class Distribution)

			Bagging	
Performance Indicator	**SVM**	**C4.5**	**SVM**	**C4.5**
Accuracy	0.718	0.687	0.718	0.739
Precision (Y[5])	0.718	0.776	0.718	0.777
Recall (Y)	1	0.793	1	0.894
F1 Measure (Y)	0.836	0.784	0.836	0.831
Precision (N)	0	0.441	0	0.559
Recall (N)	0	0.416	0	0.344
F1 Measure (N)	0	0.428	0	0.426

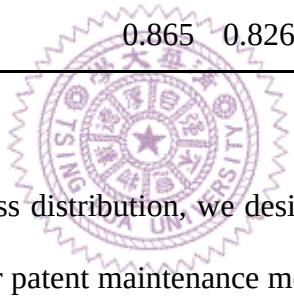
Y : N = 11642 : 4563



[5] Y denotes the “maintain” class and N represents the “abandon” class.

Table 5.4 Accuracy for Different Industries (with the Original Class Distribution)

Industry			Bagging		Maintenance
	SVM	C4.5	SVM	C4.5	Rate
Aerospace and Defense	0.845	0.779	0.845	0.84	0.845
Chemicals	0.595	0.581	0.595	0.644	0.595
Electronics, Electrical Equipment	0.714	0.646	0.714	0.697	0.714
Motor Vehicles & Parts	0.666	0.679	0.666	0.733	0.666
Network and Other Communications Equipment	0.727	0.739	0.727	0.787	0.727
Pharmaceuticals	0.628	0.616	0.628	0.67	0.628
Telecommunications	0.865	0.826	0.865	0.863	0.865



To neutralize the effects of class distribution, we design a new experiment to further analyze the effectiveness of our patent maintenance model. Specifically, we match the number of negative instances (abandoned patents) by randomly selecting the same number of positive instance (maintained patents) in each industry, and the ratio is fixed at 0.5. To avoid potential biases caused by random selection, we repeat the sampling process 10 times to create 10 "balanced" datasets and each of the "balanced" dataset is evaluated using the ten-fold cross validation strategy.

Table 5.5 shows the cross validation results in the "balanced" scenario. We can find that the accuracy, precision (Y), recall (Y), and F1 measure (Y) for the balanced scenario are lower than those achieved in the original data, because of the original dataset is skewed toward to the positive class. However, the precision (N), recall (N), and F1 measure (N) for the balanced scenario are higher than those for the original

dataset.

Table 5.5 Cross Validation Results (in the Balanced Scenario)

			Bagging	
Performance Indicator	**SVM**	**C4.5**	**SVM**	**C4.5**
Accuracy	0.617	0.596	0.618	0.645
Precision (Y)	0.636	0.597	0.637	0.649
Recall (Y)	0.546	0.594	0.547	0.63
F1 Measure (Y)	0.588	0.595	0.589	0.64
Precision (N)	0.602	0.596	0.603	0.641
Recall (N)	0.688	0.599	0.689	0.66
F1 Measure (N)	0.642	0.597	0.643	0.65

Y: N = 4563: 4563

Table 5.6 Accuracy for Different Industries (in the Balanced Scenario)

Industry			**Bagging**		**Maintenance**
	SVM	**C4.5**	**SVM**	**C4.5**	**Rate**
Aerospace and Defense	0.589	0.552	0.586	0.608	0.5
Chemicals	0.605	0.565	0.604	0.612	0.5
Electronics, Electrical Equipment	0.609	0.575	0.612	0.62	0.5
Motor Vehicles & Parts	0.616	0.626	0.618	0.664	0.5
Network and Other Communications Equipment	0.666	0.677	0.668	0.736	0.5
Pharmaceuticals	0.608	0.575	0.61	0.638	0.5
Telecommunications	0.618	0.61	0.615	0.661	0.5

Table 5.6 shows the accuracy rates of all algorithms higher than maintenance rates across all industries. The performance of the two algorithms without bagging is relatively comparable. However, with the use of bagging, C4.5 is obviously better than SVM.

5.3 Benchmark with Literature Variables

In order to demonstrate the effectiveness of our proposed patent maintenance model, we construct a benchmark model only with variables suggested by the literature. According to our literature review, the variables related to patent maintenance are shown in **Table 5.7**. In addition, the benchmark model is constructed by using the C4.5 technique with bagging algorithm.

Table 5.7 Literature Variables for Constructing the Benchmark Model

Variables
Patent scope (by IPC)
Patent scope (by USPC)
Patent family size (applications)
Patent family size (countries)
Number of backward citations
Science linkage index
Number of forward citations
Number of claims
Average length of independent claims
Length of description
Internal sequential innovation
Forward self citation rate
Backward self citation rate

Table 5.8 shows that our model achieves higher performance in most of performance indicators than the benchmark model does in the original dataset

Table 5.8 Comparative Evaluation Results (with the Original Class Distribution)

Performance Indicator	Benchmark Model	Our Model
Accuracy	0.702	0.739
Precision (Y)	0.741	0.777
Recall (Y)	0.899	0.894
F1 Measure (Y)	0.813	0.831
Precision (N)	0.436	0.559
Recall (N)	0.199	0.344
F1 Measure (N)	0.273	0.426

Y : N = 11642 : 4563

The accuracy rate for each industry attained by the benchmark model and our model are shown in **Table 5.9**. We can also find that our model gets higher accuracy rate in most of the industries except the *Telecommunications* industry (with only 0.1% difference).

Table 5.9 Comparative Evaluation Results Across Different Industries (with the Original Class Distribution)

	Benchmark Model	Our Model	Maintenance Rate
Aerospace and Defense	0.834	0.84	0.845
Chemicals	0.595	0.644	0.595
Electronics, Electrical Equipment	0.691	0.697	0.714
Motor Vehicles & Parts	0.641	0.733	0.666
Network and Other Communications Equipment	0.705	0.787	0.727
Pharmaceuticals	0.61	0.67	0.628
Telecommunications	0.864	0.863	0.865

The comparative evaluation results in the balanced scenario are illustrated in **Table 5.10** and **Table 5.11.** We can find that our model is significantly better than the benchmark model and the results confirm our conjecture that previous studies on variables for supporting patent maintenance decisions are not enough. The inclusion of our additional variables that reflect factors identified in our Delphi study improves the effectiveness of patient maintenance decisions.

Table 5.10 Comparative Evaluation Results (in the Balanced Scenario)

Performance Indicator	Benchmark Model	Our Model
Accuracy	0.558	0.645
Precision (Y)	0.559	0.649
Recall (Y)	0.554	0.63
F1 Measure (Y)	0.556	0.64
Precision (N)	0.558	0.641
Recall (N)	0.563	0.66
F1 Measure (N)	0.56	0.65

Table 5.11 Comparative Evaluation Results Across Different Industries (in the Balanced Scenario)

	Benchmark Model	Our Model	Maintenance Rate
Aerospace and Defense	0.576	0.608	0.5
Chemicals	0.576	0.612	0.5
Electronics, Electrical Equipment	0.543	0.62	0.5
Motor Vehicles & Parts	0.551	0.664	0.5
Network and Other Communications Equipment	0.583	0.736	0.5
Pharmaceuticals	0.552	0.638	0.5
Telecommunications	0.541	0.661	0.5

Chapter 6

CONCLUSION and FUTURE DIRECTIONS

6.1 Conclusion

In this research, we conduct a Delphi study to identify factors that are possible affecting patent maintenance decisions. The factors are grouped into 11 dimensions, including *business strategy*, *patent use*, *commercialization*, *market scale*, *technology transfer & licensing*, *patent family*, *patent quality*, *patent defensiveness*, *technology type*, *future R&D*, and *patent age.* The identified factors are helpful to business when facing the decision point of patent maintenance.

Besides, we propose a data-mining-based patent maintenance decision support system to support patent maintenance decisions. Our empirical evaluation results suggest that our proposed model outperforms the benchmark model (i.e., involving the variables suggested by the literature only).

6.2 Future Research Directions

First, although our study employs variables suggested in the literature and also designs some variables to reflect more patent maintenance factors. However, some factors (or dimensions), patent use for example, are hard to be operationalized by patent-related indictors. Thus, developing additional variables to reflect more patent maintenance factors identified by our Delphi study deserves more research attention.

Second, our experiment results show that the accuracy rates differ from industry to industry. For instance, the accuracy rate for the *Network and Other Communications*

Equipment industry is 73.6% (in the balanced scenario), whereas that for the *Aerospace and Defense* industry is only 60.8%. Thus, it is essential to conduct more in-depth industry analysis on how each industry evaluates their patent maintenance decisions and, if needed, develop industry-specific patent maintenance decision support system for each industry.



References

Barney, J. (2003). Comparative Patent Quality Analysis. *NACV Wh/w Pass*.

Bessen, J. (2008). The value of US patents by owner and patent characteristics. *Research Policy, 37*(5), 932-945.

Cockburn, I., Kortum, S., & Stern, S. (2002). Are all patent examiners equal? The impact of characteristics on patent statistics and litigation outcomes. *NBER WORKING PAPER SERIES*.

Crampes, C., & Langinier, C. (1998). Information disclosure in the renewal of patents. *Annales d'Economie et de Statistique*, 265-288.

Dueker, K. (1996). Trademark Law Lost in Cyberspace: Trademark Protection for Internet Addresses. *Harv. JL & Tech., 9*, 483.

Ernst, H. (2003). Patent information for strategic technology management. *World Patent Information, 25*(3), 233-242.

Griliches, Z. (1990). Patent statistics as economic indicators: a survey. *Journal of Economic literature, 28*(4), 1661-1707.

Guellec, D., & van Pottelsberghe de la Potterie, B. (2000). Applications, grants and the value of patent. *Economics Letters, 69*(1), 109-114.

Hall, B., Jaffe, A., & Trajtenberg, M. (2001). The NBER patent citation data file: Lessons, insights and methodological tools. *NBER Working Paper*.

Harhoff, D., Narin, F., Scherer, F., & Vopel, K. (1999). Citation frequency and the value of patented inventions. *Review of Economics and Statistics, 81*(3), 511-515.

Harhoff, D., Scherer, F., & Vopel, K. (2003). Citations, family size, opposition and the value of patent rights. *Research Policy, 32*(8), 1343-1363.

Jones, T., Norris, M., & Solomon, I. (2002). Strategies for maximizing value from intellectual capital in a technology-driven business. *The Licensing Journal, 22*(6), 1-7.

Lanjouw, J. O., Pakes, A., & Putnam, J. (1998). How to count patents and value intellectual property: The uses of patent renewal and application data. *Journal*

of Industrial Economics, 46(4), 405-432.

Liu, K., Arthurs, J., Cullen, J., & Alexander, R. (2008). Internal sequential innovations: How does interrelatedness affect patent renewal? *Research Policy, 37*(5), 946-953.

Maurseth, P. (2002). Lovely but dangerous: The impact of patent citations on patent duration.

Moore, K. (2005). Worthless patents. *Berkeley Technology Law Journal, 20*, 1521.

Pakes, A. (1986). Patents as options: Some estimates of the value of holding European patent stocks. *Econometrica, 54*(4), 755-784.

Pakes, A., Simpson, M., Judd, K., & Mansfield, E. (1989). Patent renewal data. *Brookings papers on economic activity. Microeconomics, 1989*, 331-410.

Schankerman, M., & Pakes, A. (1986). Estimates of the value of patent rights in European countries during the post-1950 period. *The economic journal*, 1052-1076.

Tao, J., Daniele, J., Hummel, E., Goldheim, D., & Slowinski, G. (2005). Developing an effective strategy for managing intellectual assets. *Research-Technology Management, 48*(1), 50-58.

Thomas, P. (1999). The effect of technological impact upon patent renewal decisions. *Technology Analysis & Strategic Management, 11*(2), 181-197.

Appendix

EXHIBIT 1 Original Results from Delphi Study (Traditional Chinese Version)

企業經營策略	
市場/行銷考量	未來及現有市場的銷售地區及行銷策略。
未來競爭力	***未來競爭力***[6]及商業價值考量。
經營方向變更	***並非不再有此產品，僅是企業主不再生產此類技術之產品。*** 或可考慮進行專利權讓渡、授權等。 公司結束某事業與產品線，並停止相關產品開發。
策略考量	事業群出售後，專利權如隨之移轉恐日後亦將不利於企業主本身或國內產業，且企業主本身亦不擬從事相關產品生產。 未來公司增資、上市或合併公司其無形資產所佔之比例。
專利使用	
使用情況	目前是否已使用該專利。 專利未來的"使用機率"。
製造	是否依該專利製造。
商品化程度	
是否已商品化	專利已超過五年，過去沒有專利授權或***商品化***，未來也沒有商品化計畫。 所申請專利是否有實際產品的應用或技術保護。
商品化可能性	該專利技術多為沒有價值的產品。 專利已超過五年，過去沒有專利授權或商品化，未來也沒有***商品化計畫***。

[6] 當單一句子中出現兩種以上考量專利維護之意見，我們以粗斜體字表示此句所強調之重點。

商品市場	
市場成熟度	市場型態尚未成熟，且無從預知市場成熟時間，例如 1998 年的電子商務及 1999 年的電子書，皆因此而終止專利權維護。
授權或技術移轉	
授權或技術移轉可能性	專利移轉/授權的可能性。 並非不再有此產品，僅是企業主不再生產此類技術之產品。 或可考慮***進行專利權讓渡、授權***等。 專利已超過五年，***過去沒有專利授權***或商品化，未來也沒有商品化計畫。 未來是否有專利授權或者技術移轉可能性。
是否已授權他人	該專利是否有授權他人 。
專利家族	
核心專利	該專利是否為專利家族之核心專利。
引證狀況	相關後續專利引證狀況或衍生專利概況。
是否為多國專利	是否***考量多國專利***以及年限都需維持。
專利品質	
專利品質	是否可主張專利範圍/***無程序上瑕疵/無可專利性前案影響***。
專利防禦性	
專利的防禦效率	即使該專利目前和未來都不會實施，仍應考慮是否具有”防禦專利”的價值。 防止競爭廠商或專利蟑螂的惡意訴訟行為。
保護範圍	***申請專利範圍所保護的範圍過小***，競爭對手不可能有類似的產品 例如:***申請專利範圍(Claim)權利太窄***，專利 Claim 寫法有瑕疵，專利技術容易被迴避。
迴避設計之難易度	容易經由破壞工程作迴避設計(還原工程)的技術 例如:申請專利範圍(Claim)權利太窄，***專利 Claim 寫法有瑕疵，專利技術容易被迴避。***
侵權舉證之難易度	後來發現在申請過程中存在的程序瑕疵可能導致日後主張權利權有所障礙。 ***是否可主張專利範圍***/無程序上瑕疵/無可專利性前案影響。
專利權實施之難易度	專利權實施有所困難的地區，可能是政治因素亦可能是民情風俗。

技術型態	
是否為關鍵技術	企業合併之後，經技術整合後的***關鍵技術已變更***。
前瞻性技術	目前所有產業的技術結合尚有其它關鍵技術無法突破，且非數年或十數年內可能突破，例如是智慧型機器人。
技術標準	該專利是否為技術標準中之專利。
技術之生命週期	專利技術已過時，不普及。
未來研發方向	
專利後續研發/發明的可能性	專利後續研發/發明的可能性。 該專利是否為本公司研發之重要方向之專利。 未來研發或者結合新技術可能有再申請專利機率。
專利組合可能性	未來競爭力及***商業價值考量***。
專利年限	
專利年限	專利權剩餘年限。 是否考量多國專利以及***年限***都需維持。



EXHIBIT 2 Original Results from Delphi Study (English Version)

Business Strategy	
Marketing Considerations	The marketing strategy in existing and future markets.
Future Competitiveness	***Future competitiveness***[7] and business value.
Change the Business Focus	***It's not that no longer having this product, but firms stopped producing the product of such technology.*** May consider transferring, licensing, etc. Company closes the business unit and product lines, no more products development.
Strategy	Patent may transfer after sale of the business group, and the firms would not produce related product. The proportion of intangible assets of firms with capital increase, companies listed or merged in future.
Patent Use	
Usage	Whether using the patent. Probability of future use of the patent。
Making	Whether manufactured by the patent.
Commercialization	
Whether Commercialized	With a patent life cycle of over 5 years and without patent licensing and ***patent commercialization*** in the past or plans for patent commercialization in the future. Whether the patent is protected by actual products or technology from the patent.
Possibilities of Commercialization	The patent mostly covers valueless product lines. With a patent life cycle of over 5 years and without patent licensing and patent commercialization in the past or ***plans for patent commercialization*** in the future.

[7] When the expert opinions have two or more meaning in one sentence, and we mark focus in bold and italic.

Commodity Market	
Market Maturity	Current market is yet to mature and it can't be predicted. For example, the patent in e-commerce in 1998 and e-book in 1999 was not maintained due to this reason.
Technology Transfer & Licensing	
The Possibility of Technology Transfer & Licensing	The Possibility of Transfer or Licensing. It's not that no longer having this product, but firms stopped producing the product of such technology. May consider ***transferring, licensing***, etc. With a patent life cycle of over 5 years and ***without patent licensing*** and patent commercialization in the past or plans for patent commercialization in the future. The possibility of transferring or licensing in the future.
Whether to Authorize	Whether the patent is licensed to others.
Patent Family	
Core Patent	Whether the patent is core patent in a patent family.
Forward Citation	The citations a patent received.
Multi-National Patent	Whether to maintain the ***multi-national patents*** and patent age.
Patent Quality	
Patent Quality	Is the claims valid / ***No procedural flaws*** / ***No influence from prior art***.
Defensive Patent	
Patent Defensiveness	Even if the patent will not be implemented at present or in the future, the value of its defensiveness should still be considered. Patent were used to prevent the litigation from patent troll or competition.
Claims Scope	Cause of ***narrow claims*** so the competitor would not have the similar product. E.g. ***Very narrow claims.*** Minor improvement, very easy to design around.

Difficult to Design Around	It is easy of proceeding “design around ” E.g. Very narrow claims. ***Minor improvement, very easy to design around.***
Difficult to Infringement of Proof	There are procedural flaws and it lead to the claim invalid. ***Is the claims valid*** / No procedural flaws / No influence from prior art.
Difficult to Patent Implementation	The reason that patent is hard to implement may be caused by politics or custom.
Technology Type	
Key Technology	After the merge, and ***the key technology already change***.
Innovative Technology	The technology can not break through, and it is years, maybe decades away. E.g. Intelligent robot.
Technology Standard	Whether the patent is technology standard.
Life Cycle of Technology	Patent technology phases out, not popular.
Future R&D	
Possibility of Future R&D	Possibility of future R&D. Whether the patent is in the import R&D direction. The possibility of application of the patent from future R&D and combining new technology.
Patent Portfolio	Future competitiveness and ***business value***.
Patent Age	
Patent Age	The remaining years of patent. Whether to maintain the multi-national patents and ***patent age***.

EXHIBIT 3 Detailed of Patent Maintenance/Abandonment in Our Dataset

Industry	Company	Maintained	Abandoned
Aerospace and Defense	Boeing	247	64
	United Technologies	594	91
	Honeywell Intl.	490	45
	Raytheon	117	57
	General Dynamics	53	17
Chemicals	BASF	597	464
	Dow Chemical	352	252
	Bayer	441	305
	DuPont	83	42
	Mitsubishi Chemical	97	44
	Asahi Kasei	131	21
Electronics, Electrical Equipment	Matsushita Electric Industrial	1102	181
	Samsung Electronics	1770	900
	Mitsubishi Electric	1899	868
Motor Vehicles & Parts	General Motors	536	370
	Toyota Motor	388	96
	Volkswagen	22	22
	Honda Motor	499	255
	Nissan Motor	471	184
	BMW	13	8
Network and Other Communications Equipment	Nokia	160	28
	Motorola	1654	534
	L.M. Ericsson	82	5
	Alcatel	244	178
Pharmaceuticals	Pfizer	159	97
	Johnson & Johnson	81	3
	Roche Group	1	3
	Merck	251	159
	Bristol-Myers Squibb	115	46
	Abbott Laboratories	201	97
Telecommunications	Nippon Telegraph & Telephone	172	79
	Deutsche Telekom	125	49
	France Télécom	81	13
	AT&T	1759	249

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EXHIBIT 4 Descriptive Statistics of Our Dataset

Variables	Mean	s.d.	Max.	Min.
Maintenance compatibility of technology scope (by IPC)	0.089	0.091	0.667	0
Abandonment compatibility of technology scope (by IPC)	0.078	0.097	1	0
Compatibility of technology scope (by IPC)	0.011	0.073	0.518	-1
Maintenance compatibility of technology scope (by USPC)	0.069	0.072	0.524	0
Abandonment compatibility of technology scope (by USPC)	0.056	0.075	1	0
Compatibility of technology scope (by USPC)	0.012	0.065	0.5	-1
Patent scope (by IPC)	1.869	0.956	9	1
Patent scope (by USPC)	1.838	0.94	8	1
Number of patents assignee transfer	1.369	1.467	22	0
Patent family size (applications)	8.716	17.073	777	1
Patent family size (countries)	4.376	3.832	41	1
Major market	Nominal data { 1 , 2 , 3}			
Number of backward citations	6.68	5.906	213	0
Science linkage index	1.446	3.503	74	0
Number of forward citations	10.216	16.457	520	0
Number of forward citations within 3 years	2.35	3.144	42	0
Number of forward citations within 5 years	4.59	6.557	131	0
Average number of forward citations	0.888	1.431	45.217	0
Number of claims	11.737	9.394	128	1
Number of independent claims	2.622	2.162	36	1
Average length of independent claims	179.877	99.135	2145	0
Approval time	1.786	0.799	12.28	0.38
Length of description	31206.601	27728.926	578355	3009
Originality index (by IPC)	0.346	0.281	0.913	0
Originality index (by USPC)	0.334	0.275	0.916	0

Variables	Mean	s.d.	Max.	Min.
Generality index (by IPC)	0.361	0.296	0.916	0
Generality index (by USPC)	0.358	0.286	0.918	0
Cited half life (by mean)	19.767	7.384	80.4	0
Cited half life (by median)	18.75	7.242	102.5	0
Maintenance of backward citations	1.678	0.877	4	0
Number of cross state cooperation	1.088	0.321	5	1
Internal sequential innovation	Nominal data { 0 , 1 }			
Forward self citation rate	0.129	0.233	1	0
Forward self citation rate within 3 years	0.115	0.269	1	0
Forward self citation rate within 5 years	0.123	0.242	1	0
Backward self citation rate	0.109	0.207	1	0

Instance: 16205

